Domain Join to the Closest Writable DC in an SCCM Task Sequence

Why bare metal builds join through a random domain controller when the local site only has a read-only DC, and how to fix it

Version 1.5 | 2 November 2026 | Scripts: `Set-TryNextClosestSite.ps1`, `Join-DomainClosestDC.ps1` (same folder)

The problem

During a bare metal SCCM build at a branch site, the computer account is created on a domain controller that seems random: often in another country, and different from build to build. The local DC is never used.

When the account is created on a far-away DC, the branch doesn't see it until AD replication catches up. After the reboot the machine can fail to log on or apply Group Policy, and the build slows down or fails.

The cause: the local site only has a read-only DC

- A domain join has to **create** the computer account, so it asks the DC locator for a **writable** DC.
- In this case the branch's own AD site has only a **read-only domain controller (RODC)**, which can't create accounts. Site mapping itself is fine: the machine knows which site it is in.
- With no writable DC in its site, the locator falls back to a domain-wide DNS lookup (`_ldap._tcp.dc._msdcs.<domain>`) and uses whichever writable DC answers first. That is the "random" DC.
- Fixing subnet-to-site mapping does not help, because the site is already known. The machine just has no writable DC in it.

How it was confirmed

Run these from a machine on the branch network. The output below is anonymised; the pattern is what matters.

1. Any DC: the local RODC is found

```
C:\> nltest /dsgetdc:corp.example.com
           DC: \\BRANCH-RODC01.corp.example.com
 Dc Site Name: Branch-Site
Our Site Name: Branch-Site
        Flags: GC DS LDAP KDC ... CLOSE_SITE PARTIAL_SECRET ...
```

`PARTIAL_SECRET` marks a read-only DC. `CLOSE_SITE` shows site mapping works.

2. Writable DC only: a DC on another continent

```
C:\> nltest /dsgetdc:corp.example.com /writable
           DC: \\REMOTE-DC01.corp.example.com
 Dc Site Name: Remote-Site
Our Site Name: Branch-Site
        Flags: GC DS LDAP KDC TIMESERV WRITABLE ...
```

No `CLOSE_SITE` flag. Repeating the command or a build can return a different far-away DC each time.

3. Writable, try next closest site: always the nearest hub

```
C:\> nltest /dsgetdc:corp.example.com /writable /try_next_closest_site
           DC: \\HUB-DC01.corp.example.com      (another run: HUB-DC02)
 Dc Site Name: Hub-Site
Our Site Name: Branch-Site
        Flags: GC DS LDAP KDC TIMESERV WRITABLE ...
```

Every run lands in the same nearest hub site (chosen by AD site link costs). Only the DC within that site varies, which is normal load spreading.

4. The build log agrees

In `C:\Windows\debug\netsetup.log` on a machine built with the built-in **Apply Network Settings** join:

```
NetpJoinDomain
    Domain: corp.example.com\REMOTE-DC01.corp.example.com
...
NetpJoinDomainOnDs: Passed DC 'REMOTE-DC01.corp.example.com' verified ...
NetpProvGetWindowsImageState: IMAGE_STATE_SPECIALIZE_RESEAL_TO_OOBE
```

- `IMAGE_STATE_SPECIALIZE...`: the join ran in the specialize pass of Windows Setup.
- `Domain: domain\DC`: Windows Setup had already chosen the DC (the writable lookup above) and passed it to the join. The lookup itself is in `C:\Windows\Panther\UnattendGC\setupact.log`.
- "Passed DC ... verified": when a DC is given in domain\DC form, the join goes straight to it. The fallback fix below relies on this.

Fix 1 (recommended): Try Next Closest Site

Turn on the Netlogon policy **Try Next Closest Site**. When the client's site has no suitable DC, the locator tries the next closest site by site link cost before falling back to any DC in the domain. Test 3 above shows what it picks. This keeps the built-in join, keeps ConfigMgr's own credential handling, and gives redundancy: if one hub DC is down, the other is used.

a) In the task sequence (for the build itself)

Group Policy isn't applied until after the join, so the build needs the setting written into the new OS directly. Add a **Run PowerShell Script** step in WinPE, **after Apply Operating System** and **before Setup Windows and ConfigMgr**, running `Set-TryNextClosestSite.ps1` (no parameters; it reads `OSDTargetSystemDrive`). It sets this in the offline SOFTWARE hive:

```
HKLM\SOFTWARE\Policies\Microsoft\Netlogon\Parameters
    TryNextClosestSite   REG_DWORD   1
```

The boot image needs the WinPE PowerShell component. Without it, the same thing as plain commands in a Run Command Line step:

```
cmd /c reg load HKLM\TS_Soft %OSDTargetSystemDrive%\Windows\System32\config\SOFTWARE
     && reg add HKLM\TS_Soft\Policies\Microsoft\Netlogon\Parameters
            /v TryNextClosestSite /t REG_DWORD /d 1 /f
     && reg unload HKLM\TS_Soft          (one line in the step; wrapped here)
```

b) By Group Policy (for everything after the build)

Computer Configuration > Administrative Templates > System > Net Logon > DC Locator DNS Records > Try Next Closest Site = Enabled, linked to the branch computers' OU. This keeps the setting on built machines and helps every logon at sites with only an RODC.

c) Test before rolling out

- Build one machine. In netsetup.log the `Domain:` line should now name a hub DC (HUB-DC01 or HUB-DC02 in the example), not a far-away one.
- Not yet confirmed: that the specialize-pass join reads this policy value. If netsetup.log still shows a far-away DC, use Fix 2.
- Also worth confirming the value name on a machine that already has the GPO applied: `reg query HKLM\SOFTWARE\Policies\Microsoft\Netlogon\Parameters`.

Fix 2 (fallback): join against a named writable DC

If Fix 1 doesn't work in the specialize pass, or builds must always use one particular DC, skip the built-in join and run `Join-DomainClosestDC.ps1`. It picks a writable DC from the machine's IPv4 address and calls `Add-Computer -Server domain\DC`, so no DC lookup happens at all. The trade-off: a named DC has no automatic fallback if it is down.

- **Apply Network Settings:** set to **Join a workgroup**.
- **Credentials:** create collection variables `DJUser` and `DJPassword` on the collection the task sequence is deployed to. For bare metal that is usually **All Unknown Computers**, because the machine has no device record yet. Tick **"Do not display this value in the Configuration Manager console"** for the password. It is then masked in the console, stored encrypted in policy, and not written to smsts.log.
- **Computer name:** bare metal machines need `OSDComputerName` set before Apply Windows Settings, either by prompting (an empty `OSDComputerName` collection variable on All Unknown Computers) or by a WinPE script.
- **Step:** Run PowerShell Script, execution policy Bypass, in the full OS **after Setup Windows and ConfigMgr**, then a **Restart Computer** step. Edit the configuration block at the top of the script first (domain, OU, IP prefix to DC table, default DC).
- **Check:** the script writes the DC it chose to smsts.log, and netsetup.log's Domain: line should show the same DC.

Stand-alone media builds don't download policy, so collection variables don't arrive. Use password-protected media and a hidden Set Task Sequence Variable step instead, accepting that the value is then stored in the task sequence.

After the join: the branch RODC

- The new computer account is created on a hub DC and reaches the branch RODC by normal inter-site replication. Logons straight after the build may go to the hub until then.
- Check the RODC's **Password Replication Policy**. If these computers (and their users) aren't allowed to be cached, every logon at the branch keeps going over the WAN to a writable DC.

Keeping the password out of logs

- Never pass the join password as a script parameter or a Run Command Line argument: both are written to smsts.log. If a step must take a secret as an argument, set `OSDDoNotLogCommand = TRUE` before it.
- Give the join account only delegated "create and delete computer objects" rights on the target OU.
- Limit who can deploy to All Unknown Computers: any machine that PXE boots into the task sequence can use the join account.

Order of steps

Fix 1 (recommended)	Fix 2 (fallback)
(WinPE) Partition Disk, Apply Operating System	(WinPE) Set OSDComputerName if generated; Partition Disk, Apply Operating System
(WinPE) **Set-TryNextClosestSite.ps1**	(WinPE) Apply Windows Settings; Apply Network Settings = **workgroup**; drivers
(WinPE) Apply Windows Settings; Apply Network Settings = **join domain** (as now); drivers	Setup Windows and ConfigMgr
Setup Windows and ConfigMgr (the join happens here)	(Full OS) **Join-DomainClosestDC.ps1**, then Restart Computer
(Full OS) rest of the build	(Full OS) rest of the build

Version history

- **1.5** (2 Nov 2026): Cause confirmed as a read-only-DC-only branch site (nltest /writable tests). Try Next Closest Site is now the recommended fix, with Set-TryNextClosestSite.ps1. The Add-Computer method becomes the fallback (Join-DomainClosestDC.ps1, passes domain\DC like Windows Setup does). Internal names removed for the public repo.
- **1.4**: Cause described as the initial, site-less DC lookup. Added netsetup.log analysis and the SiteName pre-seed option (dropped in 1.5: it pins laptops to one site).
- **1.0 to 1.3**: Hidden-variable credentials with Add-Computer; bare metal adjustments; -Server format; first problem statement.